E-Learning Purchase Analysis using MySQL

Objective

To analyze learner purchase behavior and course performance on an online learning platform using SQL.

The goal is to derive insights related to:

- Learner spending patterns
- Popular courses and categories
- Revenue generation
- Customer segmentation

Database Description

The dataset represents an **e-learning platform** where learners purchase digital courses.

It captures:

- Learner information (name, country)
- Course details (category, pricing)
- Purchase transactions (quantity, date)

This structure helps in analyzing **sales trends, learner behavior, and course demand**.

Tables and Structure

Learners

Column Name	Data Type	Description
Learner_ID	INT, PRIMARY KEY	A unique identifier assigned to each learner.
Full_Name	VARCHAR	Stores the complete name of the learner.
Country	VARCHAR	Represents the country of the learner.

Courses

Column Name	Data Type	Description
Course_ID	INT, PRIMARY KEY	A unique identifier for each course.
Course_Name	VARCHAR	The name/title of the course offered on the platform.
Category	VARCHAR	Defines the category or domain of the course.
Unit_Price	DECIMAL	The price of a single unit of the course.

Purchases

Column Name	Data Type	Description
Purchase_ID	INT, PRIMARY KEY	A unique identifier for each purchase transaction.
Learner_ID	INT	References the learner who made the purchase.
Course_ID	INT	References the course that was purchased.
Quantity	INT	Indicates how many units of a course were purchased in a transaction.
Purchase_Date	DATE	The date on which the purchase was made.

Tools and Technologies

- **MySQL** → Database creation & querying, Data analysis

Features Implemented

The project includes the following SQL features:

- Multi-table joins (INNER JOIN, LEFT JOIN)
- Aggregations (SUM, COUNT, DISTINCT)
- Time-based analysis using purchase_date
- Group by, Order by, Limit, Where, Having
- Ranking using ROW_NUMBER()

Analysis:

Combining Learner, Course and Purchase data:

ID	Full_Name	Country	Course_Name	Category	Price	Purchased_Date
1	Arjun Mehta	India	SQL for Beginners	Data	49.99	2025-01-02
1	Arjun Mehta	India	Python Programming	Programming	59.99	2025-01-10
1	Arjun Mehta	India	Machine Learning	AI	79.99	2025-03-30
2	Sophia Brown	USA	Power BI Dashboard	Data	69.99	2025-01-05
2	Sophia Brown	USA	SQL for Beginners	Data	49.99	2025-02-12
2	Sophia Brown	USA	Machine Learning	AI	79.99	2025-03-21
3	Oliver Smith	UK	Digital Marketing	Marketing	39.99	2025-01-07
3	Oliver Smith	UK	Machine Learning	AI	79.99	2025-02-18

Learner's Purchase Details

ID	Student_Name	Total_Courses	Total_price
1	Arjun Mehta	4	249.96
2	Sophia Brown	3	199.97
3	Oliver Smith	5	259.95
4	Liam Johnson	4	219.96
5	Rahul Verma	4	209.96
6	Ethan Wilson	4	219.96
7	Noah Fischer	5	299.95
8	Priya Nair	5	319.95
9	Emily Davis	5	309.95

Each Learner's total spending along with their country

ID	Full_Name	Country	Total_cost
1	Arjun Mehta	India	249.96
2	Sophia Brown	USA	199.97
3	Oliver Smith	UK	259.95
4	Liam Johnson	Canada	219.96
5	Rahul Verma	India	209.96
6	Ethan Wilson	Australia	219.96
7	Noah Fischer	Germany	299.95
8	Priya Nair	India	319.95
9	Emily Davis	USA	309.95
10	Lucas Martin	France	249.95

Top 3 most purchased courses based on total quantity sold

Top_3_Courses	Total_Quantity_sold
SQL for Beginners	36
Python Programming	32
Power BI Dashboard	31

Each course category's total revenue and the number of unique learners who purchased from that category

Category	Number_of_Learners	Total_Revenue
Data	35	3969.33
AI	18	1919.76
Programming	20	1919.68
Marketing	19	1199.70

Listing all learners who have purchased courses from more than one category

S_No	Name_of_the_Learner
1	Ananya Iyer
2	Antoine Moreau
3	Arjun Mehta
4	Ava Martinez
5	Benjamin Taylor
6	Charlotte Lopez
7	David Allen
8	Divya Menon
9	Emily Davis
10	Ethan Wilson
11	Felix Wagner
12	George Clark
13	Harry Williams

Courses that have not been purchased at all

Non_Purchased_Course
UI UX Design

Key Insights

- **Revenue varies significantly across categories**, indicating demand differences
- Some learners make **multiple purchases**, showing repeat customer behavior
- A few courses contribute to **majority of total sales** (top 3 courses)
- Learners from different countries show **diverse purchasing patterns**
- Certain courses have **no purchases**, highlighting potential improvement areas
- Learners purchasing from **multiple categories** indicate broader learning interest

Conclusion

- Focus marketing efforts on **high-performing categories** to maximize revenue
- Promote **low-performing or non-purchased courses** through discounts or bundling
- Target **repeat learners** with personalized recommendations
- Introduce **cross-category course bundles** to increase engagement
- Analyze country-wise trends to **localize content and pricing strategies**

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